

The technical potential for onshore wind in Wallonia

A land-eligibility and siting analysis against the Walloon regulatory framework,
and what it implies for the capacity limit used in PyPSA-Wal

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Abstract

PyPSA-Wal caps Walloon onshore wind at 6 500 MW. That number is an expert judgement taken from the PNEC wallon and from EDORA; it is not derived from a land calculation, and the model’s own calculation — the one PyPSA-Eur performs with `atlite` — was never run against Walloon law. This report closes that gap. It computes the potential from the Walloon Region’s own regulatory instruments — the plan de secteur and the Code du Développement Territorial, the *cadre de référence éolien* of 25 January 2024, the Natura 2000 network, the nature reserves, the landscape inventories, the natural-hazard maps, the aeronautical obstacle-evaluation map and the classified-site registers, all retrieved live from the Géoportail de la Wallonie. The constraints are calibrated against the 652 turbines standing in Wallonia, and turned into capacity by placing machines and parks under explicit rules rather than by multiplying area by a density.

420 km² of the Walloon Region survive every exclusion, 2.49 % of the territory. Packing machines on that land at 5.0 rotor diameters apart, with no rule about how they group, gives 11 460 MW: a land-and-wake bound. Requiring parks of at least 4 machines gives 11 084 MW, and applying the landscape rule the framework states as a result — 130° of every village’s horizon, within 4 km, free of turbines — gives the gross reference of **8 172 MW** in 453 parks. A documented allowance for the two constraint families that still have no public geometry brings the central estimate to **6.3 GW**, in a credible range of 4.6–8.2 GW.

Four results matter more than the headline number.

The answer is decided by policy rules, and each is priced. The Cadre is a circular “à valeur indicative” and the plan de secteur admits derogations for wind farms, so most of the rules that shape the answer are policy choices a permit may depart from. Section 7 prices every one of them on the reference: admitting agricultural land away from the main infrastructure network (the derogation route of CoDT art. D.IV.11) is worth +32 %, admitting every coniferous stand of the forest zone +34 %, refusing the exception that allows fewer than four machines above 3.2 MW –21 %, and applying the 2013 habitat setback –22 %.

The 4–6 km inter-distance is in the 2024 framework, as a recommendation. It is recommended between parks, “peut être réduite selon la situation paysagère locale”, does not apply along motorways, and is measured between the nearest masts. 53 % of the standing farms have a neighbour closer than 4 km. Applied as a hard rule it would bring the gross reference to 6 316 MW (4 km) or 4 892 MW (6 km); it is therefore reported as a sensitivity, not built into the reference.

BREGILAB’s 11.4 GW is a land-and-wake bound, and it cannot be reproduced from its own documentation. Its allocation packs machines on isolated pixels and applies neither the park nor the landscape rules of the Cadre; on this study’s land those two rules alone divide the land-and-wake bound by 1.40. Its documented constraint set, applied to open Walloon data, gives 25.1 GW; its published figure is 45 % of that. That its headline sits close to this study’s land-and-wake bound is a coincidence of a looser documented rule set and a tighter undocumented one.

PyPSA-Eur’s default land analysis is not usable here. It returns 19 079 MW for the Walloon Region, because it admits broad-leaved forest, ignores the plan de secteur entirely and cannot see scattered housing.

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1 The question, and the scope of the answer

1.1 The number under review

The Walloon onshore-wind capacity limit in PyPSA-Wal is a single row of `config/input_parameters_for_models.csv`:

```
potential:BEWAL:onwind:p_nom_max = 6500 MW, source “PNEC Wallon et EDORA”,
described as “potentiel technique, calculé comme la somme de ce qui est plausible d’atteindre
en 2030 comme capacité et des projets à l’étude”, with a note that EDORA estimates an
additional 3.8 GW of wind potential by 2050 on the basis of projects under environmental-
impact assessment.
```

Two things follow from that wording. It is a *deployment* expectation assembled from a pipeline of known projects, not a *land* potential; and it carries no spatial calculation that anyone can re-run or contest. For an internal scenario exercise that is acceptable. For a number reported to the administration as the ceiling on what Wallonia can host it is not: the first question is “*where does 6.5 GW fit, and under which rules?*”, and the current answer is a literature reference.

1.2 What PyPSA-Eur would have said

PyPSA-Eur ships exactly the tool needed — `atlite`’s land-eligibility analysis, driven by `scripts/determine_availability_matrix.py`. It is run for every other node in the model. For the Walloon node its output is overridden by the CSV row above, so its answer has never been examined. It is 19079 MW (19.1 GW) for the Walloon Region, on 3757 km² of admitted land.

That the override exists is defensible — Section 4 shows the default constraint set is unusable in Wallonia. That the override was never compared against a corrected calculation is the gap this report fills.

1.3 What this report answers, and what it does not

This is a land-eligibility *and siting* study. It answers: how much Walloon land may legally and physically host a wind turbine, and how many machines can stand on that land once the framework’s rules about grouping and landscape are applied? It does not answer: how much of that capacity will be permitted, financed and built, or in what year. A technical potential is an upper bound on the model’s search space, and that is precisely the role `p_nom_max` plays in PyPSA-Wal — the build-rate limit and the 2030 corridor, documented in `docs/renewable-potentials.md`, carry the deployment question.

The Walloon rules are of two kinds. Some decide whether a mast may stand at a given point at all — the zoning, the setbacks, the protected areas. Others decide how machines may be arranged — in parks, away from the horizons of villages, at a distance from other parks. The second kind is largely landscape policy, much of it indicative, and the answer depends on it more than on the first. This report therefore builds a reference on the rules as they are stated and applied, and prices every rule that is a policy or a modelling choice separately (Section 7), so that a reader who would decide one of them differently can read off the consequence.

1.4 Which Wallonia

Everything here is computed for the **administrative Walloon Region**, the union of the 21 BE3 NUTS-3 areas, 16905 km². That is the territory the Walloon rules apply to and the territory an administration will ask about.

It is not the same polygon as the model’s **BEWAL** node. PyPSA-Eur builds onshore regions by Voronoi-partitioning each country around its substations and clipping to the country shape; the partition is electrical, not administrative. The **BEWAL** polygon covers 15150 km², or 89.6 % of the Region: 2191 km² of Walloon territory — essentially the whole of western Hainaut, from Tournai

and Mouscron to Ath — sits on the Flemish side of the partition, while 435 km² of the BEWAL polygon lies outside Wallonia, mostly over the French border.

Caveat. A potential computed on one of those two regions and applied to the other is wrong regardless of how much it moves the number, and the defect affects every land-based Walloon quantity in PyPSA-Wal — ground-mounted solar and biomass among them. Fixing it means correcting the busmap, which is out of scope here and should be raised separately. Until it is, a figure from this report applied to BEWAL grants the node the resource of land the model has already given to Flanders.

2 Method

2.1 The workflow

The analysis lives in `workflow_onwind_wal/`, a self-contained Snakemake workflow. It shares nothing with the main PyPSA-Wal workflow except two read-only inputs: the region shapes and the `atlite` ERA5 cutout. It writes only inside its own directory and into `docs/onwind_potential-wallonia/`. Running it, breaking it or deleting its outputs cannot affect a model run.

```
cd workflow_onwind_wal
snakemake -c4 download # ~20 min, one-off, Walloon and OSM open data
snakemake -c4 all      # eligibility, placement, sensitivity, figures
snakemake -c4 report   # this PDF
```

The stages are:

1. **Retrieve.** Two dozen reference datasets are pulled from the ESRI-REST services of <https://geoservices.wallonie.be> (CC-BY 4.0), paginated, and written as GeoPackages with a provenance sidecar recording the service, the layer, the feature count and the retrieval date (Annex B). Feature counts are checked against the counts observed on 22 September 2026 and a deviation above 20 % raises a warning: if the Region revises a dataset, the workflow says so rather than silently changing the answer. Three inputs come from elsewhere and have their own retrieval scripts: the slope-class grid, decoded from a visualisation service (§3.5); the standing fleet, and the dual carriageways and high-speed railway lines, from OpenStreetMap (§3.3, §5).
2. **Exclusion layers.** The regulatory rules are translated into geometry, once per turbine class, because most setbacks scale with tip height or rotor diameter (Section 3). Every plan-de-secteur label the configuration names is checked against the data, so a spelling variant cannot silently drop a zone.
3. **Eligibility.** `atlite`'s `ExclusionContainer` burns every layer onto a 100 m raster in EPSG:3035 and `cutout.availabilitymatrix` aggregates the surviving share per ERA5 cell — the identical code path PyPSA-Eur uses, so the two results are comparable by construction.
4. **Calibration.** The constraint set and the siting rules are tested against the machines that are actually standing (§5).
5. **Placement.** Machines, then parks, are allocated on the eligible raster (§2.4). Every layout is checked against its own rules by an independent implementation, and the run fails if one is violated.
6. **Sensitivity.** Every policy and modelling choice is priced on its own against the reference (§7), and BREGILAB's potential is rebuilt from its own rules on the same data (§8.1).
7. **Report.** Tables, figures and every number in this text are generated. Nothing here is typed by hand.

2.2 Validation: the workflow reproduces the model’s own profile

Before any Walloon constraint is applied, the workflow has to be shown to reproduce what PyPSA-Eur itself produces. Scenario `S0_pypsa_eur` run with PyPSA-Eur’s own turbine (Vestas V112, hub 80 m) is exactly that test: same exclusion logic, same CORINE raster, same Natura 2000 raster, same ERA5 cutout.

Table 1: End-to-end validation against the profile the model actually carries. The reference is the capacity factor of `resources/walloon_2013/.../profile_adm_onwind.nc`, reported in `docs/renewable-potentials.md` §10.1. It is computed on the BEWAL polygon rather than on the Region; the two differ by less than 0.1 % in capacity factor, so the comparison is meaningful.

	this workflow	PyPSA-Wal’s own profile
full-load hours, 2013 weather year	2 442 h	2 425 h
deviation		0.7 %

Agreement here means the cutout is read correctly, the layout weighting matches PyPSA-Eur’s and the aggregation is right — so any difference in the results that follow comes from the constraint set and not from the plumbing.

2.3 Why the constraints are declared in the config, not in the code

Every exclusion is a block of `workflow_onwind_wal/config.yaml` naming the dataset, the legal instrument and the buffer, and every sensitivity case is a block naming the one rule it changes. Reviewing the constraint set — which is what the administration will want to do — requires reading one YAML file, not the Python. Changing a rule is a config edit, and Snakemake re-derives only what depends on it.

2.4 From eligible land to capacity: machines and parks are placed

The conventional step from eligible area to installable capacity is a multiplication by a capacity density in MW/km². That assumes the land arrives in blocks large enough to lay out an array. Walloon eligible land does not: it arrives as 4017 disconnected patches with a median size of 2 ha and a largest of 3.3 km², against 0.79 km² for one cell of a $5D \times 7D$ array. An area-times-density figure therefore counts land no machine could occupy. The obvious repair — screening the patches by size — fails in the other direction, because a real turbine does not have to sit inside one contiguous eligible polygon: it may stand at the edge of its patch with its rotor overhanging ineligible land, and its access track and cabling cross ineligible land as a matter of course. That is legally correct here, because every Walloon setback is measured from the *mast*.

This study therefore allocates machines explicitly, under four rules. Each is justified separately below, each is tested against the 652 machines standing in Wallonia (§5), and the last is not part of the reference.

2.4.1 Rule 1 — how close two machines may stand

A machine is accepted wherever no machine already placed, in any park, lies closer than a minimum inter-turbine distance. That distance is a wake criterion, and three readings of it are carried:

The reference is 5.0 D . Converting an area-per-turbine into an isotropic *minimum separation* forbids exactly the pairs the convention it comes from contains: in a $5D \times 7D$ array the closest two machines are $5D$ apart, not $\sqrt{35} D$. It also assumes a two-dimensional array, whereas Walloon wind farms are predominantly strings along a road, a ridge or a field boundary, where there is no downwind row and the $7D$ never binds. The physical floor is lower still — about $3D$, below which wake-induced turbulence takes a machine outside the standard IEC 61400-1 load case — and the standing Walloon fleet sits at a median of 427 m between neighbours, about $4.3 D$ at the

Table 2: Minimum inter-turbine distance, in rotor diameters, and what it gives on the reference land: before any grouping rule, and in the reference model.

case	basis	d/D	free [MW]	reference [MW]
<i>observed</i>	median nearest neighbour of the standing Walloon fleet	4.3	12 852	9 120
<i>crosswind</i>	nearest-neighbour distance of a $5D \times 7D$ array; the rule BREGILAB uses	5.0	11 460	8 172
<i>isotropic</i>	$\sqrt{5 \times 7} D$, the area-equivalent of that array	5.9	9 912	7 076

100 m-class rotor of that fleet, with a quarter of machines closer than 362 m. $5.0 D$ is therefore already the conservative end of practice.

The free allocation this rule produces on its own is the *land-and-wake bound*: 11 460 MW, 2 865 machines, an effective 27.4 MW/km^2 of eligible land. That density is far above any array density because the wake spacing between two machines on different patches is satisfied by the ineligible land between them. It depends on the order in which the land is walked — a pseudo-random order gives a result 14 % below the row-major one — and it puts a near-continuous band of machines through every corridor in the Region. It is the quantity BREGILAB reports (§8.1), and it is not a technical potential under the Walloon rules, because those rules are about how machines are arranged as much as about where a mast may stand.

2.4.2 Rule 2 — machines come in parks

The framework in force defines a park as “un projet éolien qui prévoit l’implantation de minimum quatre éoliennes” (Cadre 2024, §3.1). Parks are therefore grown from seed cells, ranked by how much eligible land surrounds them, machine by machine and nearest-first. A machine within 1.5 km of a park machine belongs to that park — the single-linkage distance used to group the standing fleet into farms — and a park is kept only if it reaches 4 machines.

Nothing else constrains the layout. There is deliberately no farm radius and no separation between farm centres: those are geometry, not rules. A model built on non-overlapping discs sterilises the land between the discs — a greedy packing of discs covers little more than half of the plane — and on this study’s land a model of farm discs 1.3 km in radius and 2.6 km apart gives about as little as the park model under the strictest reading of the inter-distance, 6 km, without any text to justify it.

The minimum of four is the definition of a park, and the same article allows it to be reduced: for repowering, for an extension of machines in operation, in a zone d’activité économique for machines above 3.2 MW, and — point 4° — for any machine above 3.2 MW “pour autant qu’il s’intègre harmonieusement dans le paysage et ne réduise pas le potentiel éolien de la zone”. The reference class is 4 MW, so point 4° applies to it. The reference reads the condition literally. Parks of at least four are grown first, and a smaller group, down to one machine, is kept only on land no such park can use, so no park is displaced. Refusing the exception and requiring four machines everywhere gives 6 480 MW (−21 %, §7). Of the machines the reference does place, 20.7 % stand in groups below four and 8.4 % are single.

A binding exclusion of single turbines is not what the texts say, and it is not the European norm.

- The 2013 framework stated a priority, not a floor: “Les parcs se composant d’un minimum de 5 éoliennes seront prioritaires”, and smaller parks “seront autorisés” provided they limit *mitage* and do not reduce the zone’s potential. The priority, it added, goes to grouping rather than to “la dispersion d’éoliennes individuelles”.
- Flanders’ current circular (OMG/2025/1) allows solitary turbines in built-up land and along large infrastructure. A minimum of three, and in principle 20 MW, applies only in open land. The previous circular had dropped an explicit clustering preference because it would foreclose

the remaining potential, and because industrial and roadside sites typically come as applications for one or two machines.

- France conditioned the purchase tariff on at least five machines in 2010 (loi n° 2010-788, art. 90) in order to fight *mitage*, and removed that condition in 2013 (loi n° 2013-312). It was a tariff rule, and it lasted less than three years.
- The Netherlands have no national minimum. Zeeland requires at least three turbines outside its designated concentration locations; Friesland restricts new solitary turbines. Measured as two or more machines within 700 m, 12 % of Dutch turbines were solitary in 2023.
- Germany sets no minimum number of turbines. Denmark’s guidance sizes the plan area to the group *or the single turbine* proposed. Ireland’s 2006 guidelines cover single turbines and groups alike. Luxembourg treats a single turbine as a case of its own, subject to the ordinary permit.

Where a binding minimum does exist, it is three rather than four, it usually lapses in industrial zones and along infrastructure, and the recent changes relax it. The argument for clustering is real — scattered machines use a zone badly, and the Pôle Aménagement du territoire said so in January 2026 when it gave an unfavourable opinion on a single turbine at Nivelles Nord (AT.26.2.AV), finding that the project would not use the wind potential of the zone and that point 4° did not apply. That is the condition the reference enforces, by refusing a small group wherever a park of four fits. It is not a reason to discard the patches that can hold one or two machines and nothing more.

The standing fleet already contains them. Grouped at the same 1.5 km, 42 of the 134 farms are single machines, 13 are pairs and 7 are triples: 89 machines, 13.7 % of the fleet, and 62 of the 134 farms. Of those machines, 56 stand in an agricultural zone, 17 in a zone d’activité économique and 16 elsewhere. The large singles that have been permitted — the Rossel turbine at Nivelles, the NRB turbine at Hauts-Sarts — stand in business zones, which point 3° of the same article opens explicitly. 86 % of the fleet is nevertheless in groups of four or more: the exception is the edge of the stock, not a different way of building it.

2.4.3 Rule 3 — every village keeps an open horizon

The framework states the landscape criterion as a result to be preserved (Cadre 2024, §3.4 §3, which repeats the 2013 text):

Pour éviter les situations d’encerclement des villages, un angle horizontal de 130° sans éoliennes est préservé sur une distance de 4 kilomètres.

It is implemented as written. Every village — a contiguous block of plan-de-secteur habitat zoning above 10 ha, 2 261 of them — carries the arcs its surrounding machines occupy, each machine subtending the angle its rotor covers from the village centre. A machine is refused if admitting it would take any village’s largest turbine-free arc below 130° within 4 km. The standing fleet respects the rule almost everywhere (§5.4), which is why it belongs in the reference.

2.4.4 Rule 4 — the inter-distance between parks, as a sensitivity

The same paragraph continues:

Sauf lorsque les éoliennes sont implantées le long des autoroutes, une interdistance minimale de 4 km (en zone de paysage à vues courtes) à 6 km (en zone de paysage à vues longues) est recommandée et peut être réduite selon la situation paysagère locale.

Five things in that sentence and in practice decide how it is treated.

1. It is in the framework in force. The 2024 circular carries it over from the 2013 framework, which called it a “référence indicative” to be taken into consideration in the light of the impact assessment; the 2024 text keeps the figures and makes the status explicit: *recommandée*, and reducible according to the local landscape.

2. It does not apply along motorways, and it depends on the landscape: 4 km where views are short, 6 km where they are long.
3. It is measured between the nearest masts of two parks. The Pôle Environnement’s 2024 opinion on a project at Chimay, for instance, finds the 4 km of the short-view Fagne-Famenne “non respectée” at 785 m, 1.2 km and 1.8 km from three neighbouring parks, among the reasons for an unfavourable opinion on a project whose permit was refused. The rule is invoked in permitting.
4. It is not what the standing fleet looks like. Measured between nearest masts, 53 % of the 134 standing farms have a neighbour closer than 4 km and 69 % one closer than 6 km; leaving out the 47 farms that stand along a motorway, 39 % are still closer than 4 km (§5.3). The reduction the text allows is the rule in practice rather than the exception.
5. The whole circular is indicative: “Un permis éolien peut s’écarter du Cadre, moyennant une motivation formelle adéquate” (§1.4).

It is implemented exactly as stated and priced as a sensitivity: a machine is refused if it would stand closer than 4 or 6 km to a machine of another park, unless both stand within 1 km of a motorway. A single uniform figure overstates the rule, since the text asks for 4 km only where views are short and allows both to be reduced; the two cases bracket it.

Finding. The land-and-wake bound is 11 460 MW. Grouping machines into parks — four or more first, then smaller groups only on the land left — costs 3 % of it and gives 11 084 MW; the open-horizon rule costs a further 26 % and gives the gross reference of 8 172 MW in 453 parks of 4.5 machines on average. The recommended inter-distance, applied on top as a hard rule, would give 6 316 MW at 4 km and 4 892 MW at 6 km. An onshore-wind potential quoted without saying which of these rules it carries cannot be compared with anything.

3 The Walloon regulatory framework

3.1 Sources of law and of practice

Five instruments define where a wind turbine may stand in Wallonia:

CoDT the *Code du Développement Territorial*, which fixes the zoning categories of the *plan de secteur*. It admits wind turbines in agricultural zones whose mast is within 1 500 m of the axis of a *principale infrastructure de communication* (PIC) or of the limit of a zone d’activité économique (art. D.II.36 §2, R.II.36-2), and in forest zones whose mast is within 750 m of a PIC, outside a recognised conservation site and outside a broad-leaved stand (D.II.37 §1, R.II.37-2). A PIC is defined at art. R.II.21-1: “les autoroutes et les routes de liaisons régionales à deux fois deux bandes de circulation”, the railway lines other than purely touristic ones, and the navigable waterways. Two further provisions matter. A *périmètre d’intérêt paysager* is not an exclusion: works there “peuvent y être autorisés pour autant qu’ils contribuent à la protection, à la gestion ou à l’aménagement du paysage” (R.II.21-7). And the permit for a wind farm, an act of public interest delivered by the fonctionnaire délégué (D.IV.22 al. 1 7° k), “peut être accordé en dérogeant au plan de secteur” (D.IV.11), under the conditions of D.IV.13.

Cadre de référence éolien the reference framework for wind development, adopted as a government circular on 25 January 2024 and applicable to applications filed from 25 April 2024; it replaced the 2013 framework. It is “une circulaire du gouvernement à valeur indicative” (§1.4). It excludes turbines from habitat, rural habitat, habitat-vert and residential-ZACC zones, from natural and park zones and from leisure zones with housing, and admits them in forest only on coniferous plantations of low biological value (§3.2 §1). It sets the mast at least **500 m plus half the total height** from those residential zones and **400 m from any dwelling**, except operators’ housing in a zone d’activité économique (§3.2 §2), where the 2013 framework asked for four times the total height from habitat zones. It defines a park as at least four machines (§3.1), states the open-horizon rule and recommends the inter-distance (§3.4 §3), and turns the infrastructure distances of 2013 into distances within which the operator must be consulted (§4).

Aeronautical servitudes the obstacle-evaluation map of the federal Direction générale Transport Aérien, published by the Region as *Hauteurs pouvant impacter l’aviation*.

Sectoral conditions and permitting practice noise limits, defence radar clearances, and the protection of the Humain radio-astronomy station and the Wideumont weather radar.

The Walloon favourable-zone cartography the 2013 map of *zones favorables* and its 2022 update by SPW and Gembloux Agro-Bio Tech, which is the Region’s own operational translation of the rules into GIS layers. Its constraint tables are the template followed here.

3.2 The integral exclusions, and how each is implemented

Table 3 lists the integral exclusions of the Walloon favourable-zone methodology, what this workflow does with each, and where the implementation differs.

3.3 The zoning rule that dominates: the agricultural corridor

The CoDT does not treat agricultural land as uniformly available. It admits a turbine there only within 1.5 km of the axis of a PIC or of the limit of a zone d’activité économique, and in coniferous forest only within 750 m of a PIC. This is a *concentration* rule — keep wind development along the existing structuring infrastructure rather than scattered across the open landscape — and it has no analogue in PyPSA-Eur, whose own treatment works the other way, by *excluding* a 1 km band around transport infrastructure.

What counts as a PIC decides how much land the rule admits. The plan-de-secteur road layer is not the PIC network: besides 729 km of motorways it draws some 6 128 km of roads in total, most of them two-lane *routes de liaison*. Taking every one of them as a PIC puts a 1.5 km corridor around almost every main road of the Region and admits a large share of the agricultural land the rule is meant to keep free. The road part of the network is built here from the plan-de-secteur motorways and the OpenStreetMap dual carriageways — motorways, and trunk, primary and secondary roads drawn as one-way pairs with a limit of 70 km h⁻¹ or more, 3 171 km of carriageway — which is what “deux fois deux bandes de circulation” means on the ground. Reading every plan-de-secteur road as a PIC instead would move the reference by +23 % (§7).

The corridor governs a permit *in conformity* with the plan de secteur. A wind farm can also be permitted in derogation of it, and that route is used: 79 % of the standing machines stand on land the zoning admits with the CoDT’s PIC network, 86 % with every road taken as a PIC. The derogation route — agricultural land admitted wherever the other rules allow — is the largest single policy lever in this study and is priced in §7.

3.4 Setbacks for the three turbine classes

The turbine classes mirror the three *scénarios de base* of the 2022 favourable-zone update (total tip height 150, 180 and 210 m) so the two studies can be read side by side. The reference class is NREL 2020ATB 4.0 MW (tip 185 m): rotor 150 m, tip height 185 m, giving a habitat-zone setback of 592 m under the 2024 framework.

Caveat. Read as $4H$ on the total height — the rule of the 2013 framework — the habitat setback would be 740 m for the reference class instead of 592 m. The 2024 rule is used because it is the one in force; the 2013 rule is priced in §7 and available in the config (`setbacks.habitat_zone.default: cdr2013`).

3.5 The three constraint families added from data that had to be rebuilt

Three of the families in Table 3 are not published as ready-made exclusion polygons. They are reconstructed here, and each reconstruction is a hypothesis in its own right.

Table 3: Walloon integral exclusions and their implementation. **Red** marks a constraint carried as a documented allowance rather than as geometry (§6.4).

category	Walloon rule	implementation here
technical incompatibility	water bodies, extraction zones, steep slopes	plan-de-secteur <i>Plan d'eau</i> and extraction classes excluded; <i>versants</i> $> 30^\circ$; and the $\geq 7\%$ slope criterion of the favourable-zone methodology from the ERRUISSOL slope grid (§3.5)
plan de secteur	broad-leaved forest excluded; coniferous forest admissible only within 750 m of a PIC; agricultural land only within 1.5 km of a PIC or of a zone d'activité économique; <i>espaces verts, naturelle, parc</i> excluded	a positive list on the zoning classes of the <i>Zones d'affectation</i> layer, every label checked against the data; the PIC network of R.II.21-1 (§3.3); the broad-leaved/coniferous split from CORINE, which the plan de secteur does not carry (§3.6)
infrastructure safety	2013 exclusion distances, 2024 consultation distances: roads $1.5 \times$ blade length; railways 50 m, high-speed lines 190 m; HV lines $1.5 \times$ rotor diameter	buffers on the plan-de-secteur networks and the OSM high-speed lines; for the reference turbine 112 m on roads, 50 m on railways, 190 m on 312 km of high-speed track and 225 m on HV lines
hazard zones	flood-prone areas, landslide risk, karst, close water-capture protection perimeters	EU Floods-Directive overflow extent (extreme return period), <i>contraintes physiques liées aux éboulements, zones de prévention rapprochée IIa</i> , and the <i>contrainte forte</i> level of the karst map (131 of 449 polygons; the moderate and weak levels call for a geotechnical study, which a foundation gets anyway)
civil aviation and national defence	airport control zones, obstacle-limitation surfaces, radar and beacon interference zones, military airspace	the DGTA obstacle-evaluation map, restricted to the two innermost aerodrome rings after calibration against the standing fleet (§3.5, §5); skeyes and Defence red zones are not published
built heritage	classified sites, protection zones, UNESCO buffers	<i>sites classés, ensembles architecturaux, zones de protection</i> and the UNESCO buffer zones, from the Region's registers
natural heritage	nature reserves, forest reserves, wetlands of biological interest, underground cavities of scientific interest, Natura 2000	all five implemented from the FAUNE-FLORE services
biodiversity	high-priority ornithological zones	allowance — the DEMNA priority-bird layer is not public
landscape	landscape preservation perimeters; open horizon of each village; inter-distance between parks	plan-de-secteur <i>périmètres d'intérêt paysager</i> and the ADESA inventory excluded (a conditional overlay and a non-binding inventory, both strongly avoided by the fleet, §5); the open horizon in the park model and the inter-distance as a sensitivity (§2.4)
living environment	habitat zones of the plan de secteur, $500\text{ m} + H/2$; dwellings, 400 m	cdr2024 rule, 592 m for the reference turbine, around habitat, rural-habitat, ZACC and communal-stake zones; 400 m around the ICAR address points, except the 18 721 that lie in a zone d'activité économique
scientific	Humain radio-astronomy station, Wideumont weather radar	reconstructed circles at the published coordinates (§3.5)
partial constraints	medium-priority bird zones, bat-interest zones, migration corridors, the principal ecological structure, radar study zones at 8–16 km, each granted a 25 % success rate	allowance — representing them needs a permit-level model rather than an area calculation

Aeronautical servitudes. The Region publishes *Hauteurs pouvant impacter l’aviation* (AOEM), which gives, for every point of Wallonia, the height above which the Direction générale Transport Aérien must be consulted. The six classes — 0, 15, 25, 35, 50 and 60 m — tile the territory exactly. Four of them are nested rings around the aerodromes (Liège, Charleroi, Beauvechain, Florennes, Chièvres, Saint-Hubert, Bertrix, Spa and the smaller fields); two are the territory-wide notification background, one polygon of 8 045 km² at 25 m and one of 1 556 km² at 60 m. Any wind turbine exceeds every one of those thresholds, so the map is a consultation duty and not, on its face, an exclusion. Which of the aerodrome rings is a genuine no-go is settled empirically in §5: the 0 m and 15 m rings are strongly avoided by the standing fleet and are excluded here; the 35 m and 50 m rings carry *more* turbines per square kilometre than Wallonia as a whole and are not.

The 7 % slope criterion. The Walloon favourable-zone methodology treats slopes above 7 % as incompatible with the crane pad and access track of a large machine. It is a technical criterion of that methodology, not a rule of the CoDT or of the Cadre. The Region publishes the slope grid — `ERRUISOL_CLASSE_PENTE`, per cent slope derived from the 10 m `ERRUISSOL` terrain model — but only as a visualisation service, with no feature query and no coverage download. What the service does expose is an image export that renders an exact bounding box into an exact number of pixels with nearest-neighbour resampling and a seven-colour palette, one colour per slope class. Decoding those colours recovers the class grid losslessly, and the 7 % threshold falls exactly on a class boundary, so no interpolation is involved; so do the 10 % and 15 % thresholds priced in §7. The grid is rendered at 25 m and a 100 m cell is excluded when the *majority* of its samples are at or above the threshold — that is, when most of the hectare a turbine needs is too steep, not when a single steep pixel touches it. On that rule 37.5 % of the Walloon Region is excluded, and the reconstruction passes two independent checks: 99 % of the cells the Region’s separate $> 30^\circ$ vector layer flags are inside it, and its north/south split (19.5 % of the northern third against 49.3 % of the southern third) is the Hesbaye–Ardenne gradient. Among land that passes every other rule, the standing fleet avoids it strongly (§5).

Radar and radio astronomy. skeyes and Defence do not publish radar-constraint geometry, and the AOEM map covers obstacle-limitation surfaces, not radar interference. Three installations whose protection perimeters fall inside Wallonia and are not already covered by an aerodrome ring are reconstructed as circles at their published coordinates: the Wideumont C-band weather radar (5 km, WMO/OPERA guidance), the Humain radio-astronomy station (3 km) and the Bertem primary/secondary surveillance radar (10 km, of which only the southern arc reaches Walloon Brabant). Together they remove 9 km², so the approximation is not load-bearing. The skeyes and Defence *red zones* themselves are a different matter: they are published neither as geometry nor as rules, BREGILAB applies them (§8.1), and they are the one family whose absence here could matter.

3.6 The forest species split

The CoDT excludes a mast standing in a broad-leaved stand. The plan de secteur records forest zoning but not species, so the split has to come from a land-cover product. This study uses CORINE grid codes 23/24/25 at 250 m, 2006 vintage, counting mixed stands as broad-leaved — the conservative choice, and the same source PyPSA-Eur uses.

Two better sources were identified and neither can be used:

- **CARTOFOR**, the continuous mapping of the Walloon forest by Gembloux Agro-Bio Tech under the Accord-Cadre de recherche forestière, has a 2016 three-class product distinguishing broad-leaved from coniferous stands. Its own documentation states that the cartographic products are undergoing validation and will be made available on the Region’s portal in due course; they are not published today, and no service exposes them.

- **WALOUS 2018**, the Walloon land-cover grid, does carry explicit *Feuillus* and *Résineux* classes at a far finer resolution than CORINE, and would be the correct implementation of the CoDT rule — including for the broad-leaved woodlots that sit inside agricultural zones, which the present implementation misses. Its service answers point queries but returns a blank image to the bulk export that the slope grid accepts, so it cannot be retrieved as a grid. Obtaining it is an administrative step.

Caveat. The stake of the species split, under the CoDT’s distance, is bounded. It only decides the status of plan-de-secteur forest zones lying within 750 m of a PIC, which is a small share of a Region whose eligible land is overwhelmingly agricultural; and the direction of the error from using CORINE is not knowable without the better data, because a coarse raster both admits broad-leaved cells and rejects coniferous ones. Dropping the distance, which the next subsection prices, makes the split load-bearing.

3.7 Wind in forest, in Wallonia and elsewhere

Two Walloon instruments do not say the same thing. The CoDT admits a mast in a forest zone only within 750 m of a PIC, outside a recognised conservation site and outside a broad-leaved stand (D.II.37 §1 al. 6, R.II.37-2). The 2024 framework, which is indicative, admits turbines “sur les plantations de résineux pauvres en biodiversité” and does not repeat the distance (§3.2 §1). The reference follows the CoDT, because it is the statute and because the framework cannot open land the code closes. The framework’s wording is priced on its own.

It is not a hypothetical. 10 standing machines sit in a plan-de-secteur forest zone, in 4 farms. 1 of them stands where the CoDT’s distance and the conifer reading both admit a mast; 7 stand on coniferous stands with the distance dropped, so most of the forest machines were permitted beyond 750 m of a PIC. 10 of the 10 fall inside the forest zone once broad-leaved stands are admitted too. The practice is a derogation from the distance, on conifers, and it is ten machines, not a programme.

Elsewhere the same question is answered more openly, and almost nowhere by a ban.

- Germany had 2 533 turbines and 7 406 MW on forest land at the end of 2024, about 9 % of the machines and 12 % of the onshore capacity (Fachagentur Wind und Solar, 2025). Forest sites are open in Baden-Württemberg, Bavaria, Brandenburg, Hessen, Rhineland-Palatinate and Saarland. North Rhine-Westphalia allows them on calamity areas and in municipalities whose forest cover exceeds 20 %. Hamburg, Bremen and Schleswig-Holstein do not. A conversion of forest still needs a permit under §9 of the federal forest act, and nature reserves and national parks stay closed. The turbines stand in commercial forest, not in ancient broad-leaved woodland.
- France has no ban. Clearing woodland needs an authorisation under art. L.341-3 of the Code forestier, with compensatory planting, and public forest is cleared only with the agreement of the Office national des forêts. Projects in forest are built; they are not the bulk of the stock.
- Scotland presumes against woodland removal (Control of Woodland Removal Policy, 2009) and allows it, with compensatory planting, where it facilitates renewable energy. Commercial conifer plantations are a routine host of wind farms. The guidance that implements the policy is written for wind farms first.

Admitting every coniferous stand of the Walloon forest zone, which is the framework’s wording without the CoDT’s distance, gives 10 988 MW (+34 %). Admitting the whole forest zone, broad-leaved stands included, gives 14 024 MW (+72 %). The first is a genuine disagreement between the two instruments in force, and it is the reading closest to the few forest machines that exist; it is large because the Ardennes are windy, not because the standing fleet is there. It is a policy sensitivity, not the reference. The second is contrary to both instruments. It is kept as an upper bound, so that the cost of treating forest as ordinary land — which is what PyPSA-Eur does — stays visible. It is not a plausible rule.

Both readings are on the dashboard, as the three positions of one control: the CoDT’s distance, every coniferous stand, and the whole forest zone. The middle position is worth a control because it moves the answer by +34 %. The third is worth keeping beside it so that a reader can see what “all forest” would mean, and can see that it is not the law.

4 PyPSA-Eur’s default constraint set is not applicable in Wallonia

PyPSA-Eur’s onshore-wind land analysis (`config/config.default.yaml`, section `renewable.onwind`) admits CORINE grid codes 12–29, 31 and 32, excludes a 1 km buffer around codes 1–6, excludes Natura 2000, and assumes 3 MW km^{−2}. Five features of that specification make it unsuitable here.

1. **It admits all forest.** Codes 23, 24 and 25 are broad-leaved, coniferous and mixed forest. Wallonia is roughly one third forest, and the forest is concentrated in the Ardennes — the windiest part of the Region. Walloon law excludes broad-leaved stands outright and admits coniferous stands only near a PIC. This single difference moves the answer more than any other.
2. **It cannot see scattered housing.** CORINE’s minimum mapping unit is 25 ha. Walloon *habitat dispersé* — isolated farms and ribbon development — is invisible to it, and it is exactly what the 400 m rule protects. The 1 km buffer around CORINE urban codes is not a substitute: it is generous where settlements are large and blind where they are not.
3. **It has no zoning.** The plan de secteur is the legal instrument that decides admissibility in Wallonia, and CORINE is a land-*cover* product that knows nothing about it.
4. **Its buffer around transport infrastructure has the wrong sign.** PyPSA-Eur excludes 1 km around roads and railways; the Walloon rule requires agricultural sites to be *within* 1.5 km of the main ones.
5. **The CORINE raster shipped in the data bundle is 250 m, 2006 vintage** (`g250-clc06_V18_5.tif`). At 250 m a single pixel is 6.25 ha; at the scale of a 400 m setback that is a coarse instrument, and the land cover is twenty years old.

Finding. PyPSA-Eur’s default land analysis admits 3 757 km² of Wallonia and returns 19 079 MW. That figure should not be quoted as a Walloon technical potential, and the decision to override it in `input_parameters_for_models.csv` was correct. What was missing was a corrected calculation to override it *with*.

5 Calibration against the standing fleet

A land-eligibility study is a claim about where a machine may be built. The cheapest way to find out whether the claim is calibrated is to point it at the machines that exist. The Region does not publish turbine coordinates; OpenStreetMap does, and its coverage of the Belgian fleet is essentially complete. 652 turbines are mapped inside the Walloon Region. They are used only to test the constraint set and the siting rules — never as an input to the potential.

5.1 Which constraints the fleet avoids

For each constraint layer, Table 4 reports the share of the Region it covers, the share of the standing fleet it covers, and the ratio of the two. A ratio well below one says that developers and the permitting authority treat the constraint as real; a ratio at or above one says that they do not, whatever the layer is called.

That ratio is reported twice. Against the whole Region, it confounds each layer with all the others: steep slopes lie mostly in forest that the zoning excludes anyway, so a slope layer’s Region-wide ratio mostly measures the forest. The *conditional* ratio (columns marked *) counts only the land, and only the machines, that pass every other layer of the set; it is the one that tests

the layer. The two size-dependent setbacks are left out of the conditions, because most standing machines are smaller than the reference class.

Table 4: Avoidance of each constraint layer by the 652 turbines standing in Wallonia, reference turbine class. *: conditional on every other layer except the two size-dependent setbacks.

constraint layer	Region [%]	fleet [%]	ratio	land* [%]	fleet* [%]	ratio*
plan de secteur, ineligible zoning	62.9	20.7	0.33	53.1	20.2	0.38
setback from habitat zones	62.0	23.3	0.38	—	—	—
setback from scattered dwellings	62.1	12.3	0.20	—	—	—
road, rail and HV-line setbacks	13.9	11.5	0.83	19.7	12.1	0.61
Natura 2000, reserves, wetlands, caves	13.3	0.2	0.01	3.8	0.0	0.00
landscape perimeters (PIP / ADESA)	31.8	5.5	0.17	19.1	4.4	0.23
of which plan-de-secteur PIP	18.7	0.8	0.04	5.7	0.2	0.04
of which ADESA inventory	21.1	5.1	0.24	16.6	4.1	0.25
flood, karst, landslide, water capture	10.9	3.5	0.32	9.1	4.6	0.51
aeronautical servitudes	7.5	1.5	0.20	12.1	1.5	0.12
slope $\geq 7\%$	37.5	5.5	0.15	24.6	3.7	0.15
classified sites and protection zones	1.9	0.2	0.08	0.5	0.0	0.00
radar and radio astronomy	0.7	0.0	0.00	0.8	0.0	0.00

Finding. Every layer of the reference set is avoided by the standing fleet, and the landscape, nature, aviation and slope layers strongly so: conditional ratios of 0.04 for the plan-de-secteur landscape perimeters, 0.25 for the ADESA inventory, 0.12 for the aeronautical rings and 0.15 for the slope criterion. The layers the fleet avoids least are the infrastructure distances — conditional ratio 0.61 — which the 2024 framework itself turned into consultation distances, and the hazard zones (0.51).

The same test settles two choices. The **35 m and 50 m aeronautical rings** cover 2.9 % and 8.5 % of the Region but carry 5.5 % and 13.6 % of the standing fleet (ratios 1.88 and 1.60); they are consultation rings, not exclusions, and are not applied. The **runoff component of the flood hazard** maps a thin-sheet phenomenon that does not preclude a foundation; the overflow extent at the extreme return period is used instead.

Caveat. Only 44 % of the standing fleet survives the full constraint set — 287 of 652 machines. Most of the rest stands inside the setbacks of the reference class: the habitat setback is sized for a 185 m machine, and much of the fleet is smaller and was permitted under the 2002 and 2013 frameworks. The zoning is the other large term: 79 % of the fleet stands where the zoning admits a turbine with the CoDT’s PIC network, so about one machine in five was permitted by derogation or before the present code. The potential computed here is therefore conservative relative to what has actually been permitted.

5.2 What the addresses around the fleet are

134 standing machines have an ICAR address point within 400 m. For 65 of them the nearest such address lies in a zone d’activité économique, and for 36 in an agricultural zone. The first are businesses, or the operators’ housing the 2024 rule exempts; they are not dwellings in the sense of the rule, and the reference drops the 18 721 addresses that lie in those zones. The second are farms, most of them dwellings. Everywhere else every address is kept, so shops, schools and barns still count as dwellings: the setback remains mildly conservative.

5.3 How the fleet is grouped

Grouped by single linkage at 1.5 km, the 652 machines form 134 farms of 4.9 machines on average; 72 of those farms have four machines or more and hold 86 % of the fleet; the largest has 27.

Measured the way the Region measures its inter-distance — between the nearest masts of two farms — the distance from each farm to its nearest neighbour has a p10 of 1 584 m, a p25 of 2 042 m and a median of 3 798 m: 53 % of the farms are closer than 4 km to another and 69 % closer than 6 km. 47 farms stand entirely within 1 km of a motorway, where the inter-distance does not apply; of the others, 39 % are still closer than 4 km to a neighbour. Only 34 % of the machines stand within 1 km of a motorway (37 % within 1.5 km), so the motorway exemption concerns a third of the fleet, not most of it.

5.4 The open horizon is observed

Of the 2 261 villages, 880 have a standing machine within 4 km, and only 6 of them (0.7 %) have less than 130° of their horizon free of turbines; the p1 of the largest free arc is 152°. The rule is respected in practice, and the implementation used in the placement model reproduces that — it is the same code.

6 Results

6.1 How much land each rule removes

Scenarios S1–S6 are cumulative: each adds layers to the previous one, so the difference between two consecutive rows is the cost of the rules that were added. S0 is *not* part of that sequence — it is PyPSA-Eur’s own specification, a different constraint set rather than a subset of S1 — and it is shown apart in the table and in grey in the figure.

Finding. The Walloon plan de secteur on its own leaves 6 266 km² admissible. Adding the setbacks of the cadre de référence — 592 m from habitat zones and 400 m from every dwelling — removes 5 292 km², or 84 % of it, in a single step. The Walloon onshore-wind land envelope is governed by settlement pattern, not by nature protection and not by the wind.

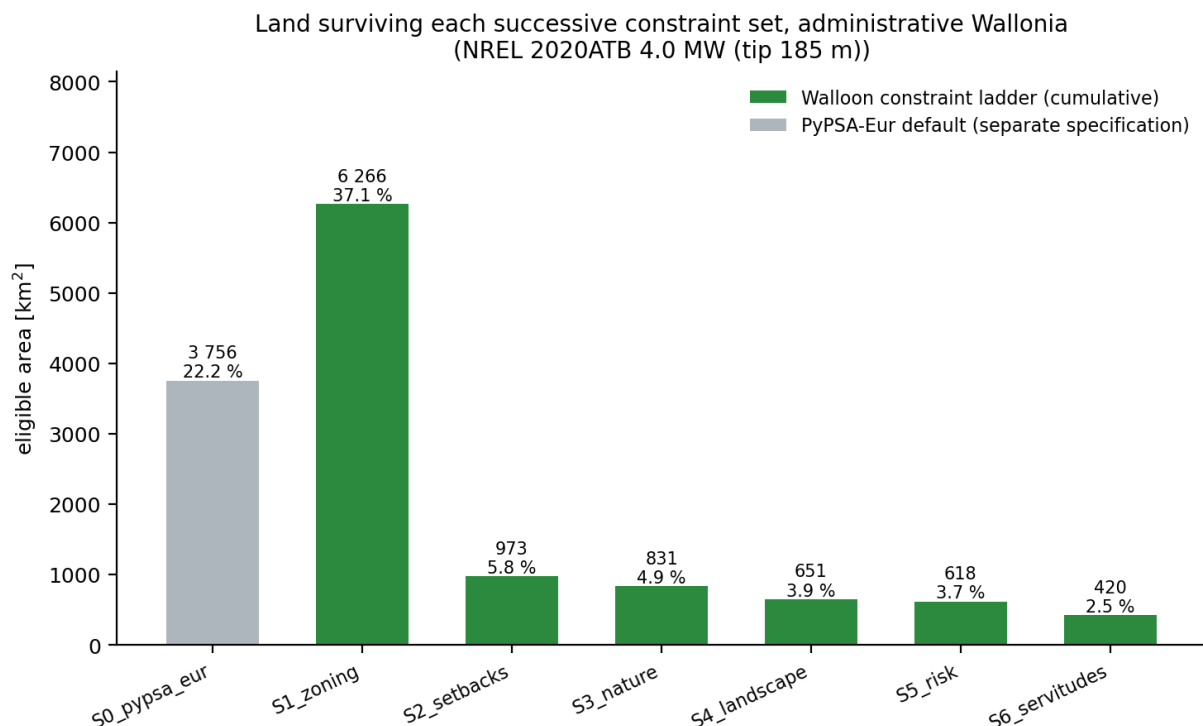


Figure 1: Land surviving each successive constraint set, for the reference turbine class.

Table 5: Eligible area after each constraint set, administrative Walloon Region, reference turbine class.

scen.	constraint set	area [km ²]	share [%]	removed [km ²]
S0 pypsa eur	PyPSA-Eur default (CORINE + Natura 2000)	3 756	22.22	n/a
S1 zoning	+ plan de secteur zoning (CoDT)	6 266	37.06	10 640
S2 setbacks	+ dwellings and infrastructure setbacks (CDR 2024)	973	5.76	5 292
S3 nature	+ Natura 2000, reserves, wetlands, caves	831	4.92	142
S4 landscape	+ landscape protection perimeters (PIP / ADESA)	651	3.85	180
S5 risk	+ flood, karst, landslide, water-capture protection	618	3.66	33
S6 servitudes	+ aeronautical servitudes, slope $\geq 7\%$, classified sites, radar	420	2.49	198

6.2 What the servitude families cost

Table 6 isolates the four families of §3.5 and the heritage registers, each measured on its own against the land S5 leaves. Marginal costs do not add up to the combined cost, because the families overlap each other and overlap what S5 has already removed.

Table 6: Cost of each constraint family added on top of S5, administrative Walloon Region, reference turbine class. S5 leaves 621 km².

constraint family	removes [km ²]	[%]	leaves [km ²]
aeronautical servitudes (DGTA obstacle map)	71	11.4	550
slope $\geq 7\%$ (ERRUISSOL, 10 m DTM)	128	20.6	493
classified sites, protection zones, UNESCO buffers	2	0.4	618
weather radar, radio astronomy, Bertem SSR	9	1.4	612
all four together	203	32.7	418

Finding. The four servitude families are together worth 33 % of the surviving land. The 7 % slope criterion alone removes 128 km² (20.6 %) and the aeronautical servitudes 71 km² (11.4 %); classified sites and radar perimeters are worth 2 and 9 km² respectively and could have been left as an allowance without harm.

6.3 The eligible land, and what fits on it

420 km² survive every implemented constraint, 2.49 % of the Region, in 4 017 patches of median size 2 ha. Only 32 % of that area sits in patches large enough to hold one cell of a $5D \times 7D$ array and 1.6 % in patches large enough for four — which is why this study places machines rather than multiplying by a density (§2.4).

Finding. The reference layout places 2 043 machines in 453 parks — median 2 machines, largest 55 — at an effective 19.55 MW/km² of eligible land, and 8 172 MW. It passes an independent check of its own rules: no two machines closer than 762 m, every park of 4 machines or more, and no village with less than 130.0° of free horizon.

The full allocation results, for every turbine class and spacing, are in Annex A.

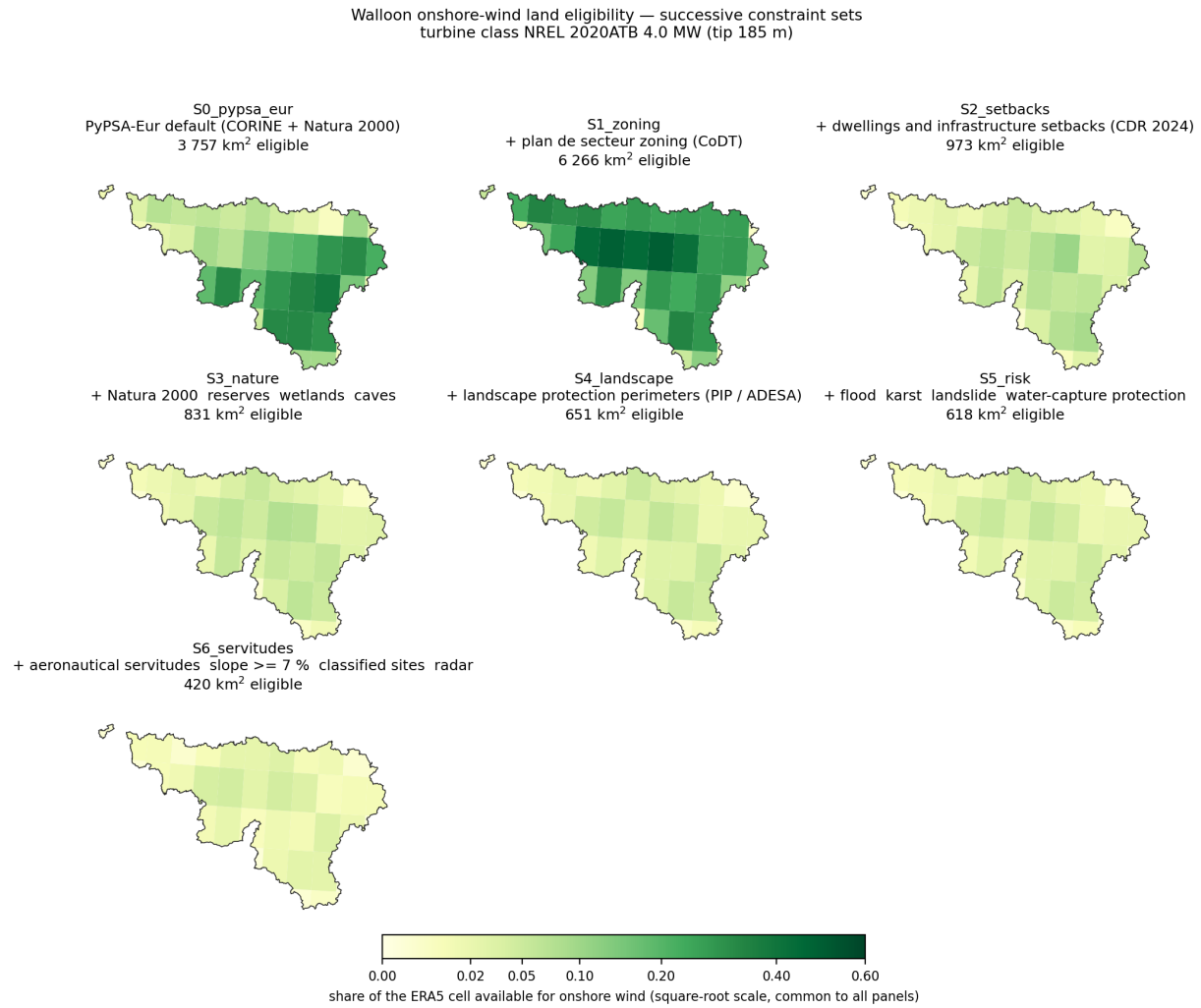


Figure 2: Spatial effect of each constraint set. The shading is the share of each ERA5 cell that survives; the ERA5 grid is 0.3° , so these panels show *where* the resource is, not the parcel-level pattern.

Table 7: What each placement rule costs, reference turbine class and $5.0 D$ spacing. The last two rows are the inter-distance the Cadre recommends, applied as a hard rule: a sensitivity, not the reference.

	parks	machines	P [MW]
free allocation, 750 m between machines	—	2 865	11 460
+ parks of 4 or more, 1.5 km linkage	611	2 771	11 084
+ open horizon, 130° within 4 km (gross reference)	453	2 043	8 172
<i>sensitivity</i> : + 4 km between parks, motorways exempt	284	1 579	6 316
<i>sensitivity</i> : + 6 km between parks, motorways exempt	193	1 223	4 892

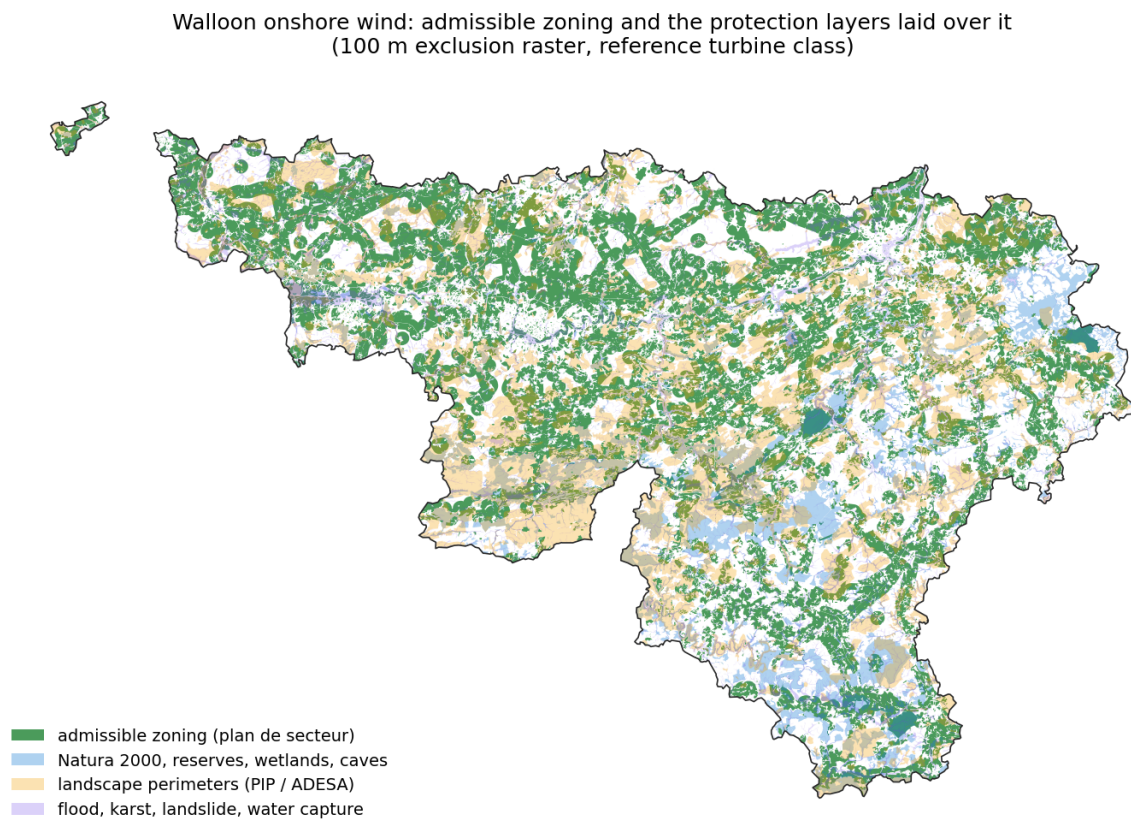


Figure 3: The admissible plan-de-secteur envelope with the protection layers laid over it, at the 100 m resolution of the exclusion raster.

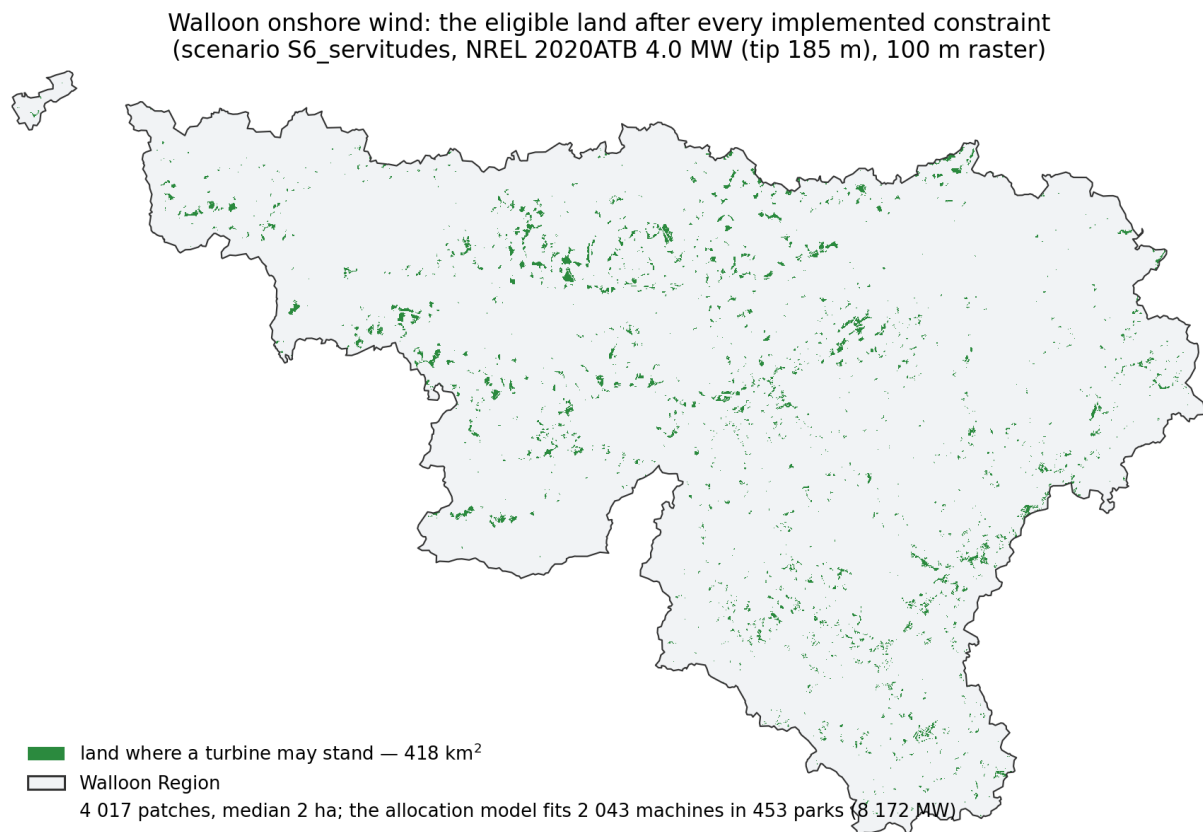


Figure 4: The land that survives *every* implemented constraint, at 100m resolution. This is the raster the machines are placed on, and it is written out as `results/eligible_land_{turbine}.tif` so it can be opened in QGIS and checked parcel by parcel; the placed layouts are in `results/placement_{turbine}.gpkg`.

6.4 The residual allowance

Two families of the Walloon method have no public geometry and both reduce the potential. They are represented by a multiplicative allowance on the reference, with a central value and a bracket: **Priority ornithological zones, 10 % (bracket 0–20 %)**. The DEMNA high-priority bird layer is an integral exclusion of the favourable-zone method and is not published. It was built largely from concentrations of sensitive species, which in Wallonia coincide to a large extent with the Natura 2000 bird SPAs and the wetlands already excluded here; the incremental area is the non-Natura remainder. The central value is a judgement, not a measurement, and it is stated as one.

Partial constraints, 15 % (bracket 0–30 %). The method distinguishes integral from *partial* exclusions — medium-priority bird zones, bat-interest zones, migration corridors, the principal ecological structure, radar study zones between 8 and 16 km — and grants turbines falling in them a 25 % success rate. Their combined footprint is unknown; the central estimate assumes they cover a fifth of the surviving land, so the expected loss is $0.20 \times 0.75 = 15\%$.

Together they leave 76 % of the reference in the central case.

6.5 The answer

Table 8: From eligible land to a technical potential, administrative Walloon Region, reference turbine class at 5.0 D spacing.

	P [MW]	
eligible land, all implemented constraints		420 km ²
× capacity density, 5.08 MW/km ²	2 136	<i>convention only</i>
free allocation, 750 m between machines	11 460	<i>land and wake bound</i>
+ parks of 4 or more	11 084	
+ open horizon, 130° within 4 km	8 172	<i>gross reference</i>
× residual allowance, central	6 252	central estimate
credible technical potential	4 576–8 172	
<i>if the recommended 4–6 km inter-distance were applied as a rule</i>	3 742–4 832	
PyPSA-Wal cap today	6 500	
standing fleet, end 2024	1 528	

Finding. The credible gross technical potential for onshore wind in the administrative Walloon Region is **4.6–8.2 GW**, central estimate 6.3 GW, on the reference turbine class and the framework in force as it is applied. The range is the residual-allowance bracket of §6.4 applied to the reference case; it is not statistical, and it does not include the policy choices of §7, which move the answer further in both directions. Applying the recommended inter-distance as a binding rule would give 3 742–4 832 MW, and choosing a machine spacing of 4.3 D or 5.9 D instead of 5.0 D would move the gross reference between 9 120 and 7 076 MW.

6.6 Full-load hours and energy

The capacity factor is computed on the 2013 ERA5/SARAH-3 cutout with the same layout weighting PyPSA-Eur uses. For the reference case it is 3 260 h, which with the reference capacity gives 26.64 TWh/yr gross and 20.38 TWh/yr after the residual allowance — against the 6.2 TWh/yr of the PNEC/PACE 2030 target.

Those full-load hours are high, and the reason is the machine, not the method. The reference class is an NREL 2020 ATB 4 MW turbine with a 150 m rotor — a specific power of 226 W m^{−2}, against 305 W m^{−2} for the V112 PyPSA-Eur uses by default — on a 110 m hub instead of 80 m.

Low-specific-power machines on tall towers are exactly what raises full-load hours, and it is why the same workflow returns 2 467 h when the V112 is substituted into the same eligible land. The V112 number is the one that reconciles with §2.2 and with the model’s own resource.

Caveat. These are *gross* new-turbine full-load hours: no wake, availability or electrical losses are applied, which is PyPSA-Eur’s convention for `p_max_pu` and therefore the convention PyPSA-Wal needs. They are not comparable with the 1 885 h TIMES-WAL uses for the blended existing fleet, nor with what the standing Walloon fleet delivers; that discrepancy is analysed in `docs/renewable-potentials.md` §10. BREGILAB applies a calibrated performance ratio of 0.69 to the same kind of figure (§8.1), which is the right correction for a study reporting delivered energy.

Caveat. Only the 2013 weather year is available on this machine. The Walloon onshore-wind resource in 2013 is about 5% *better* than in 2010, the year the main model runs use (`docs/renewable-potentials.md` §10.1). Capacity is unaffected — it depends only on the land — but the energy column is optimistic by roughly that margin relative to a 2010 run.

Table 9: Reference scenario S6_servitudes across turbine classes and capacity densities. *derived* is $P_{\text{nom}}/(5D \times 7D)$; *pypsa eur* is PyPSA-Eur’s 3 MW km^{−2}; *dense* is a $4D \times 6D$ upper bound. These are the area-times-density figures, retained for comparison with studies reported on that convention.

turbine	density	[MW km ^{−2}]	area [km ²]	P [MW]	FLH [h]	E [TW h]
V112 / 3.0 MW (tip 136 m)	dense	8.00	463	3 705	2 467	9.1
	derived	6.83	463	3 164	2 467	7.8
	pypsa eur	3.00	463	1 389	2 467	3.4
NREL 2020ATB 4.0 MW (tip 185 m)	dense	8.00	420	3 364	3 260	11.0
	derived	5.08	420	2 136	3 260	7.0
	pypsa eur	3.00	420	1 261	3 260	4.1
NREL 2020ATB 5.5 MW (tip 208 m)	dense	8.00	398	3 185	3 334	10.6
	derived	5.13	398	2 043	3 334	6.8
	pypsa eur	3.00	398	1 194	3 334	4.0

Turbine class barely moves the energy, only the capacity. Across the three classes the energy yield per square kilometre of eligible land is 16.6–17.1 GWh/km², a spread of a few per cent, while the reference model on the same rules returns 8 172–9 493 MW across the three classes. That is the expected physics: at fixed rotor-diameter spacing the swept area per square kilometre is a constant, so a low-specific-power machine trades megawatts for full-load hours at roughly constant energy. The choice of turbine class is therefore close to irrelevant for what the model needs from this study — annual energy — and decisive for the headline capacity number, which is a reason to report both.

7 What the answer depends on

A technical potential is a statement about what the rules permit, and some of the rules are policy. The Cadre de référence is a circular “à valeur indicative” that a permit may depart from with adequate reasons, and the plan de secteur itself admits derogations for wind farms. A potential that silently builds in one reading of a policy rule answers a political question without saying so. Table 10 therefore prices every choice the reference rests on, one at a time: each row is the reference with exactly that one change, through the same eligibility code and the same checked placement model.

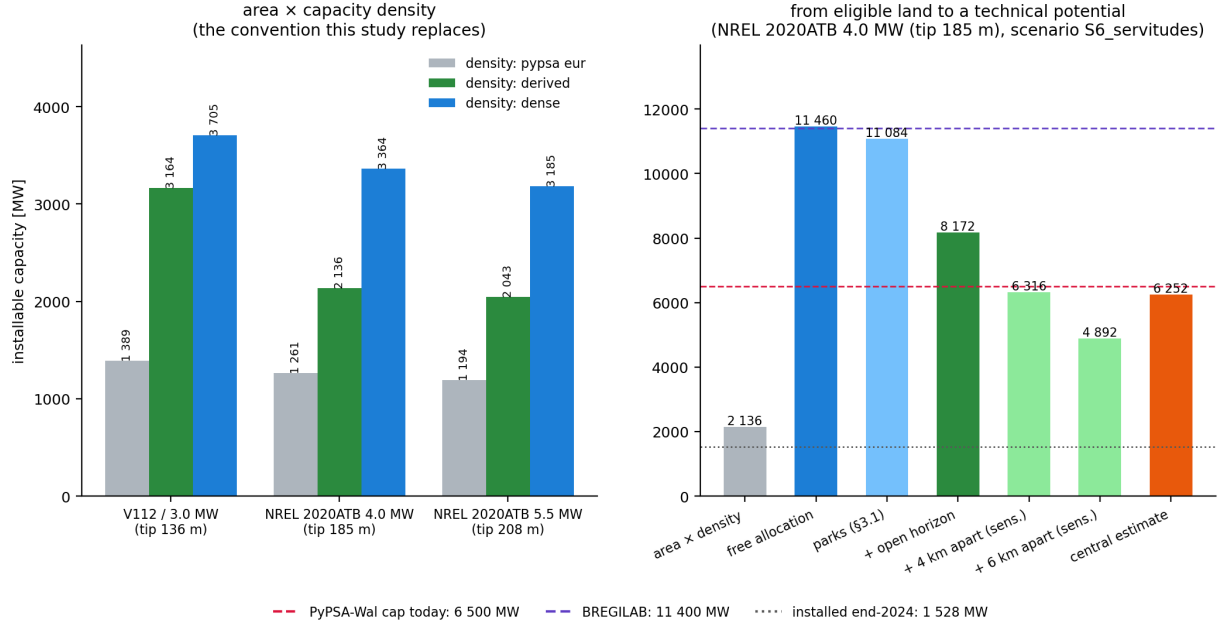


Figure 5: Left: installable capacity in the reference scenario by turbine class and capacity density. Right: the ladder from eligible land to the technical potential.

Table 10: The reference with one change at a time (reference turbine class, 5.0 D). *policy*: a rule a government sets and a permit can depart from; *judgement*: a modelling choice of this study; *misreading*: a reading of the CoDT this study does not follow; *combined*: several changes at once.

change from the reference	kind	area [km ²]	P [MW]	Δ
reference (2024 framework as applied)		418	8 172	–
inter-distance 4 km between parks (short views), motorways exempt	policy	418	6 316	–23 %
inter-distance 6 km between parks (long views), motorways exempt	policy	418	4 892	–40 %
no open-horizon rule	policy	418	11 084	+36 %
parks of at least four, without the 2024 exception above 3.2 MW	policy	418	6 480	–21 %
no agricultural corridor (derogation, CoDT D.IV.11)	policy	667	10 772	+32 %
2013 habitat setback, 4 x tip height	policy	279	6 408	–22 %
ADESA perimeters not excluded	judgement	480	8 992	+10 %
no landscape perimeter excluded	judgement	507	9 356	+14 %
slope criterion at 10 %	judgement	487	8 712	+7 %
slope criterion at 15 %	judgement	524	8 996	+10 %
no aeronautical ring excluded	judgement	484	8 980	+10 %
every coniferous stand of the forest zone admitted (Cadre 2024 wording, no PIC distance)	policy	615	10 988	+34 %
the whole forest zone admitted, broad-leaved stands included	policy	954	14 024	+72 %
every plan-de-secteur road taken as a PIC (a misreading of R.II.21-1)	misreading	557	10 028	+23 %
no agricultural corridor and every coniferous stand, open horizon kept	combined	862	13 368	+64 %
6 km inter-distance and parks of at least four without exception	combined	418	4 336	–47 %

Policy rules. Five rules decide the answer, and none of them is about the land in the physical sense.

- *The agricultural corridor* (CoDT R.II.36-2) keeps turbines within 1.5 km of the main infrastructure network. It is the rule for a permit in conformity with the plan de secteur; the derogation route of D.IV.11 is used for about one standing machine in five (§5). Admitting agricultural land wherever the other rules allow gives 10 772 MW (+32 %), on 667 km² of eligible land.
- *The forest zone.* The reference follows the CoDT: coniferous stands within 750 m of a PIC only. The framework’s wording, every coniferous stand and no distance, gives 10 988 MW (+34 %); the whole forest zone, broad-leaved stands included, gives 14 024 MW (+72 %). The first is the disagreement between the two instruments; the second is an upper bound (§3.7).
- *The park minimum.* The reference applies point 4° of §3.1: parks of four first, then smaller groups on the land left. Requiring four machines everywhere, which is the definition without its exception, gives 6 480 MW (−21 %).
- *The inter-distance between parks,* recommended and reducible: 6 316 MW at 4 km (−23 %) and 4 892 MW at 6 km (−40 %).
- *The habitat setback:* the 2013 rule of four times the tip height instead of the 2024 rule gives 6 408 MW (−22 %). It is no longer in force. It stays in this table because the published studies that still use it, BREGILAB among them, have to be read against it. It is not a choice on the dashboard.

The open-horizon rule belongs to the same family, but it is stated as a result to be preserved and the standing fleet observes it; dropping it would give 11 084 MW (+36 %). Lifting the corridor and admitting every coniferous stand, with the open horizon kept, gives 13 368 MW; tightening the inter-distance to 6 km and requiring four machines everywhere gives 4 336 MW. *The policy envelope around the gross reference is therefore 4 336–13 368 MW*, wider than the residual-allowance bracket and wider than any modelling choice.

Modelling choices. Three exclusions are this study’s judgement, and the fleet supports each of them (§5): the ADESA landscape inventory, which has no legal force (without it, 8 992 MW, +10 %; without any landscape perimeter, 9 356 MW, +14 %); the slope criterion (8 712 MW at 10 %, 8 996 MW at 15 %); and the two inner aeronautical rings (8 980 MW without them).

A misreading. Taking every plan-de-secteur road as a PIC — the reading BREGILAB appears to share (§8.1) — would give 10 028 MW (+23 %). That is not a policy choice but a reading of R.II.21-1, which is explicit, and it is not followed.

Finding. The reference is the framework in force as it is applied: the CoDT corridor and its forest distance, parks of four with the exception for machines above 3.2 MW on the land a park cannot use, the open horizon, no binding inter-distance. The largest movements are the 6 km inter-distance (−40 %), the open horizon (+36 % if dropped), every coniferous stand without the CoDT’s distance (+34 %) and the agricultural corridor (+32 %). The 2013 habitat setback (−22 %) and refusing the park exception (−21 %) each move it by about a fifth. A reader who wants a potential under a different policy should take it from Table 10 rather than from the headline.

8 Against the other estimates

8.1 BREGILAB / VITO Dynamic Energy Atlas

The other published bottom-up estimate of the Walloon onshore-wind potential is the BREGILAB project’s, computed with VITO’s Dynamic Energy Atlas (Clymans et al., 2022, WP3, §5.2.2 and Table 22). Its Table 22 gives Wallonia 1 010 MW current and 10 390 MW additional onshore wind; the 11.4 GW quoted in its running text is their sum, that is, a *gross* potential on the same

Table 11: This study against the figures currently used or published. All capacities are gross, i.e. they include the standing fleet.

source	basis	P [MW]	E [TWh]
PyPSA-Wal today (input_parameters_for_models.csv)	PNEC wallon / EDORA, expert judgement	6 500	21.19
PyPSA-Eur default land eligibility	this study, scenario S0, area x 3 MW/km ²	19 079	61.62
BREGILAB / VITO Dynamic Energy Atlas	Clymans et al. 2022, V112 at 5 D, free allocation, gross	11 400	18.20
BREGILAB’s rules on this study’s data	this study, emulation of the WTN scenario, V112 at 5 D	25 087	–
Walloon constraint set, area x density	this study, S6_servitudes, 5.08 MW/km ²	2 136	6.96
Walloon constraint set, free allocation	this study, S6_servitudes, 750 m between machines, no grouping rule	11 460	37.36
Walloon constraint set, parks of 4 or more	this study, S6_servitudes, no landscape rule	11 084	36.13
Walloon constraint set, parks and open horizon	this study, S6_servitudes, gross reference	8 172	26.64
Walloon constraint set, machines placed, residual allowance	this study, central allowance for the families with no geometry	6 252	20.38
SPW/Gembloux favourable-zone update 2022, ‘180 m’ case	site-by-site simulation, additional potential only	–	3.96
Installed fleet, end 2024	EDORA / Renouvelle	1 528	–
PNEC / PACE 2030 onshore-wind target	Walloon Government	–	6.20

convention as `p_nom_max`. It is $\times 1.4$ this study’s gross reference and $\times 1.8$ its central estimate. The difference can be traced, and most of it is not about the land.

What BREGILAB computes. It delineates “available space” on a 100 m grid from the 2013 cadre de référence, the CoDT and the 2009 ICEDD study — its WTN scenario, Tables 20–21 — and then runs an allocation algorithm that packs in as many Vestas V112 (3.3 MW, hub 94 m) as will fit at a minimum spacing of $5D = 560$ m. Its own Figure 21 shows machines allocated on isolated pixels of available space: the algorithm is a free allocation in the sense of §2.4. Constraints are combined by a weighted overlay with an exclusion threshold of 0.5; the Walloon landscape rules — the 130° open horizon and the inter-distance, quoted verbatim in its Table 20 — enter only as a layer of “protected views (4 km) around urban settlements” with a weight of 0.4, which by construction excludes nothing on its own. Production is corrected by a performance ratio of 0.69, calibrated against Elia’s measured output — which is why its reported availability factor for the Walloon potential is 1 586 h and not the 2 298 h implied gross, a figure close to what this workflow gets for the same machine, and a useful independent check that the two resource models agree.

Its rules on open data. To see whether the land explains the number, the workflow rebuilds BREGILAB’s WTN constraint set on this study’s data: the 2013 habitat setback of four times the 150 m tip height, 400 m from dwellings, its negative advice limits around roads, railways, waterways and HV lines, 2 km around the nuclear sites, 560 m around the standing fleet, Natura 2000 and the reserves, forest and green zoning, heritage, high-risk karst, the aeronautical no-go rings standing in for its aviation red zones, and its 1.5 km corridor along the “road, rail and waterway network”. It admits 1 034 km², and a free allocation of its machine on that land, with the standing fleet kept, gives 25 087 MW. BREGILAB publishes 11 400 MW, 45 % of that. The available space its Figure 20 maps for Wallonia is visibly sparser than its own rules produce on open data; the layers

that could account for the difference — skeyes and Defence red zones, SEVESO and pipeline buffers, cadastral residential parcels instead of address points, undeveloped industrial parcels — are listed in its tables but not published, so its figure cannot be reproduced from its documentation.

The bridge. Table 12 starts from its rules on this study’s data and adds this study’s rules one at a time, all with BREGILAB’s machine and spacing, then switches to the reference machine and to the park and landscape rules.

Table 12: Bridge from BREGILAB’s rules to this study’s answer. Upper block: free allocation of BREGILAB’s V112 (3.3 MW, 560 m), each row adding one of this study’s rules. Lower block: the reference machine and this study’s placement rules.

step	area [km ²]	<i>P</i> [MW]
<i>V112, 3.3 MW, 560 m apart, free allocation</i>		
BREGILAB’s rules (WTN), this study’s data	1 034	25 087
+ landscape perimeters (plan de secteur and ADESA)	804	19 932
+ flood, landslide and water-capture zones	764	19 470
+ slope ≥ 7 %	574	16 678
+ radar and radio-astronomy perimeters	566	16 460
+ corridor restricted to the CoDT’s PIC network	425	12 576
+ 2024 habitat setback (500 m + H/2) instead of 4H	453	13 309
<i>reference machine, this study’s rules</i>		
NREL 2020ATB 4.0 MW (tip 185 m), 5 D, free allocation	420	11 460
+ parks of 4 or more, smaller groups on the land left		11 084
+ open horizon, 130° within 4 km (gross reference)		8 172
× residual allowance (central estimate)		6 252
BREGILAB, as published (Table 22)		11 400

Why BREGILAB’s number is not a technical potential under Walloon rules. Three things separate it from this study’s answer, and they do not point the same way.

1. **It is a land-and-wake bound.** It places machines one pixel at a time, forms no parks and applies neither the park definition nor the open-horizon rule of the Cadre. On this study’s land those two rules take the land-and-wake bound from 11 460 MW to 8 172 MW. This is the main reason its headline is high, and it is a matter of method, not of data.
2. **Its documented constraint set is looser than the Walloon rules.** It does not exclude the landscape perimeters, slopes or flood zones, and its corridor follows every road rather than the CoDT’s PIC network. On its own machine, this study’s rules take its constraint set from 25 087 MW to 13 309 MW (Table 12).
3. **Its implemented constraint set is tighter than its documentation.** Its published figure is only 45 % of what its own rules give on open data, which is what the undocumented layers remove.

The second and third effects pull in opposite directions and happen to cancel: BREGILAB’s 11 400 MW sits close to this study’s land-and-wake bound of 11 460 MW. The agreement is a coincidence, not a confirmation, and the factor $\times 1.4$ between BREGILAB and this study’s gross reference is the first effect: BREGILAB counts the machines that fit, this study counts the machines the Walloon rules allow to be arranged in parks.

What was adopted from it. Three things, and together they changed the answer of this study more than any new data layer did:

1. **Allocating machines instead of multiplying area by a density.** This is BREGILAB’s method, it is better than an area calculation on fragmented land, and it removes the need for any patch-size screen.

2. **Graded rather than binary constraints.** BREGILAB’s weighting of consultation zones is the idea behind this study’s separate treatment of the aeronautical rings and behind the residual allowance for partial constraints.
3. **Validating against measured reality.** BREGILAB calibrates its production against Elia. This study cannot do that for a potential, but the same instinct produced the fleet calibration of §5.

What was not adopted, and why. BREGILAB’s performance ratio of 0.69 is a calibration factor that absorbs both physical losses and the bias of its own wind model against Elia’s measurements. Applying it to a PyPSA-Eur profile would double-count, because PyPSA-Wal expects a gross `p_max_pu`. Its weighted-overlay treatment of soft constraints was also not reproduced literally: with only two soft families left unmapped here, an explicit allowance with a stated bracket (§6.4) is more transparent than a hidden weight — and a weight of 0.4 on the landscape rules is, in effect, a decision not to apply them.

8.2 The 2022 SPW/Gembloux favourable-zone update

That study reports 175 km² of *zones favorables strictes* and 272 km² of *zones favorables élargies* for its “180 m” case, and 3 959 GWh/yr of *additional* production from 457 potential turbines, cut down from 1 323 simulated ones by landscape inter-distance and encirclement criteria.

Three differences are structural:

- It applies the priority bird zones, which this study represents only by an allowance, and it applies the landscape inter-distance and the encirclement examination — the criteria that cut its own simulation from 1 323 turbines to 457 — where this study keeps the open horizon and treats the inter-distance as a sensitivity.
- It counts only *new* capacity, excluding the footprint of the standing fleet, whereas this study reports a gross potential.
- It applies a wind-resource threshold; this study applies none.

Adding the standing fleet back to its 457 turbines gives of order 4 GW on a comparable gross basis, which is where this study lands once the inter-distance is applied as a rule (3 742–4 832 MW) — the reading of the landscape rules that the 2022 update made.

8.3 The model’s own cap

6 500 MW against this study’s gross reference of 8 172 MW is a factor of 0.80 — the cap is, to within a percent, the gross reference — and against its central estimate of 6 252 MW a factor of 1.0. Two qualifications matter.

First, the cap was constructed as a deployment expectation — what is plausible by 2030 plus the project pipeline — and this report produces a land-and-siting envelope. That the two land in the same range is a coincidence of construction, not a confirmation.

Second, whether the cap is inside the envelope depends on the policy choices of §7: it is reachable under the reference rules only if none of the unmapped constraint families binds, it is out of reach under a binding inter-distance, and it is comfortably inside the envelope on the derogation route for agricultural land.

8.4 Three independent cross-checks

The conclusion of §6.5 rests on a chain — constraint set, eligible raster, machine allocation, park allocation, residual allowance — and an error anywhere in it would move the answer. Three checks test the end of the chain against things that are known independently of it.

The layout obeys its own rules. Every layout in this report is checked against the rules it claims to satisfy by an implementation separate from the one that produced it — spacing, park size, inter-distance and open horizon — and the workflow fails if one is broken (§6.5).

The parks. The reference places 453 parks against the 134 farms that exist, that is, it says Wallonia could host somewhat more parks than it has, most of them larger. The median park has 2 machines against a fleet average of 4.9, and the tail is long — the largest has 55 machines against 27 in the fleet — because nothing but the open horizon limits a park’s extent. The inter-distance sensitivities remove parks rather than shrink them: at 6 km the model keeps 193 parks.

The resulting deployment density, against a country that has built one. Germany ended 2024 with 63.55 GW of onshore wind in 28 717 machines on 357 600 km² — a deployed density of 178 kW/km² and 80.3 machines per 1 000 km². Wallonia today carries 90 kW/km², 0.51 times that. This study’s central estimate corresponds to 370 kW/km², 2.08 times Germany’s current deployed density, and the top of its range to 2.72 times.

Finding. A technical potential that lands at 2.08 times the density Germany has *already built* is in a credible place: above what Wallonia has done, not far above what a comparable country has done, and well below the densities reached in Schleswig-Holstein or Lower Saxony.

Caveat. The allocations are greedy, not optimal: parks are grown from the best seeds first and never revisited. An exact packing would fit somewhat more, so every placement figure is a slight under-estimate of what the rules permit — in the opposite direction to every other approximation in this report.

9 Hypotheses

Every choice below is a config entry, every one can be varied, and each is stated with the direction in which it moves the result; the policy and judgement choices are priced in Table 10.

Region. The administrative Walloon Region, not the model’s BEWAL node (§1.4). *A BEWAL figure would be about 10 % lower.*

Capacity comes from placing machines and parks, not from multiplying area by a density (§2.4). *The density convention would give 2 136 MW for reasons that have nothing to do with the rules.*

Minimum inter-turbine distance $5.0 D = 750$ m, between any two machines, the nearest-neighbour distance of a $5D \times 7D$ array and the rule VITO uses. *The standing Walloon fleet sits at $4.3 D$ (9 120 MW); the area-equivalent $\sqrt{5} \times 7 D$ would give 7 076 MW.*

Parks grown at the fleet’s 1.5 km linkage: at least 4 machines first, then smaller groups only on the land no such park can use (Cadre 2024, §3.1 4°), no radius and no centre separation. *Without the exception, 6 480 MW.*

Open horizon 130° free within 4 km of every village above 10 ha, as the Cadre states it and as the fleet observes it. *26 % of the parks figure.*

Inter-distance between parks not binding in the reference; 4 and 6 km between nearest masts, motorways exempt, as sensitivities. *6 316 and 4 892 MW.*

Agricultural corridor 1.5 km from the PIC network of R.II.21-1 or a zone d’activité économique. *The derogation route gives 10 772 MW.*

Habitat setback the 2024 framework, $500 + H/2$. *The 2013 rule ($4H$) gives 6 408 MW.*

Dwellings 400 m around every ICAR address point outside the zones d’activité économique. The address layer includes non-residential buildings, so this is mildly conservative; the 2022 SPW study makes the same approximation.

Infrastructure distances roads $1.5 \times$ blade length, railways 50 m, high-speed lines 190 m, HV lines $1.5 D$: the exclusion distances of 2013, now consultation distances. *Weakly avoided by the fleet; conservative.*

Aeronautical servitudes the AOEM rings at 0 and 15 m only, after the fleet calibration of §5. *Without them, 8 980 MW.*

Slope $\geq 7\%$ on the ERRUISSOL 10 m grid, majority rule over the 25 m samples in each 100 m cell. *At 15 %, 8 996 MW.*

Landscape perimeters the plan-de-secteur PIP and the ADESA inventory, both excluded. *Without ADESA, 8 992 MW.*

Karst the *contrainte forte* level only.

Radar and radio astronomy three circles at published coordinates with generic separation distances (§3.5). *Worth 9 km^2 , so not load-bearing.* The skeyes and Defence red zones are not represented at all. *Reduces, by an unknown amount.*

Broad-leaved / coniferous split CORINE at 250 m, mixed stands counted as broad-leaved; CARTOFOR and WALOUS would both be better and neither is retrievable (§3.6). *Direction unknown.*

Residual allowance 10 % for priority bird zones and 15 % for partial constraints, both bracketed (§6.4). *Both reduce; the brackets are carried into the headline range.*

Runoff flood hazard excluded from the flood layer; the overflow extent at the extreme return period is used instead.

Geometry generalisation heavy polygon layers are requested with 25 m server-side generalisation — a quarter of the exclusion raster cell, so it cannot change the aggregate.

Existing wind farms are not excluded. The result is a *gross* potential: the standing 1 528 MW occupies land that is counted as available. This is the right convention for `p_nom_max`, which caps the total fleet, not additions.

Repowering is not a separate term. A land-and-siting study measures the envelope, not the turnover of the fleet inside it. EDORA’s additional 3.8 GW by 2050 explicitly includes repowering, which is part of why that number and this one are not the same kind of quantity.

No grid-connection or economic screen. Distance to the HV network, connection cost and wind-speed profitability thresholds are not applied. *All reduce.*

Energy is gross. No wake, availability or electrical losses, matching PyPSA-Eur’s `p_max_pu` convention. *A delivered-energy figure would be roughly 15 % lower; BREGILAB’s calibrated performance ratio is 0.69.*

Weather year 2013. Capacity is unaffected; energy is about 5 % optimistic relative to the 2010 year the model runs use.

10 What this means for `p_nom_max`

1. **The cap can be documented rather than cited.** This report gives a gross reference of 8 172 MW and a central estimate of 6 252 MW under the framework in force as it is applied, and prices every policy rule around them. Replacing the CSV’s “PNEC wallon et EDORA” with that derivation — and with the policy reading it carries — removes the main weakness of the current number.
2. **If a single value is wanted, 6 252 MW is the defensible one** for a potential under present rules. 6 500 MW is above it by a factor of 1.0; it is reachable within the present rules only if the unmapped constraint families do not bind, and within the policy envelope of §7 on the derogation route for agricultural land.
3. **State the policy reading with the number.** A potential under a binding 4–6 km inter-distance (3 742–4 832 MW) and one on the derogation route (10 772 MW gross) are both defensible readings of Walloon rules; they differ by more than the residual-allowance bracket, and PyPSA-Wal should say which one it uses.
4. **Settle the region question before quoting anything externally.** A Region figure applied

to the BEWAL node overstates the node by about 10 % (§1.4).

5. **Decide what `p_nom_max` is meant to be.** PyPSA-Wal already carries a build-rate limit and a 2030 corridor (`docs/renewable-potentials.md` §3–4) to represent deployment. If those are doing their job, `p_nom_max` should be the technical potential and the present value is the wrong kind of number in the right range. That is a modelling decision, not a GIS one.

11 What would sharpen the number further

None of these is a gap that invalidates the result; each would narrow the range.

1. **Ask SPW Énergie for the DEMNA priority-bird layer and the partial-constraint layers.** They exist and the favourable-zone cartography uses them. They would replace the residual allowance, which is the widest single term in the range, with geometry.
2. **Ask skeyes and Defence for their wind-turbine red zones.** They are the one integral constraint family this study cannot represent and BREGILAB can, and they are the most likely explanation of the gap between BREGILAB’s rules and its published figure.
3. **Ask SPW Territoire how the inter-distance and the derogation route are applied.** Where the 4 km and 6 km figures are reduced in practice, and how often agricultural land away from the PIC network is permitted, are the two policy questions the answer depends on most; a map of short- and long-view landscapes would let the inter-distance be applied at the right value in each.
4. **Request WALOUS 2018 as a grid, or CARTOFOR when it is published.** Either would replace a 2006 250 m raster with a Walloon product and would implement the CoDT broad-leaved rule where it currently is not applied at all (§3.6).
5. **Repeat on the 2010 weather year** once its cutout is available, to put the energy column on the same footing as the model runs.

A Full results

B Data provenance

All layers below were retrieved from the ESRI-REST services of the Géoportail de la Wallonie (<https://geoservices.wallonie.be>) under the CC-BY 4.0 licence.

Additional inputs: the slope-class grid `SOL_SOUS_SOL/ERRUISOL_CLASSE_PENTE`, decoded from the Region’s visualisation service (§3.5); the standing wind fleet, the dual carriageways and the high-speed railway lines from OpenStreetMap via the Overpass API (ODbL 1.0); ERA5/SARAH-3 `atlite` cutout for 2013 (Copernicus); CORINE Land Cover 2006 at 250 m, PyPSA-Eur data bundle; NUTS-3 shapes and the PyPSA-Wal onshore regions, PyPSA-Eur resources; turbine power curves from the `atlite` library (Vestas, NREL 2020 ATB).

Legal texts: Circulaire du Gouvernement relative au cadre de référence éolien, 25 January 2024; Cadre de référence pour l’implantation d’éoliennes en Région wallonne, 21 February 2013, as amended on 11 July 2013; Code du Développement Territorial, decretal and regulatory parts, articles D.II.28, D.II.36, D.II.37, D.IV.11, D.IV.13, D.IV.22, R.II.21-1, R.II.21-7, R.II.36-2 and R.II.37-2; Pôle Environnement, opinion ENV.24.106.AV on a wind project at Chimay-Baileux (2024); Pôle Aménagement du territoire, opinion AT.26.2.AV on a single turbine at Nivelles Nord (16 January 2026). Comparative instruments cited in §2.4 and §3.7: Flanders, circular OMG/2025/1; France, loi n° 2010-788, art. 90, and loi n° 2013-312; Zeeland, *omgevingsverordening*, art. 18.17; Fachagentur Wind und Solar, *Entwicklung der Windenergienutzung auf Forstflächen*, 10th ed. (2025); Code forestier, art. L.341-3; Scottish Government, *Control of Woodland Removal Policy* (2009).

Table 13: All scenarios and turbine classes at the reference capacity density, administrative Walloon Region.

scenario	turbine	area [km ²]	P [MW]	FLH [h]	E [TWh]
S0 pypsa eur	V112 / 3.0 MW (tip 136 m)	3 757	25 668	2 442	62.67
S1 zoning	V112 / 3.0 MW (tip 136 m)	6 266	42 814	2 464	105.51
S2 setbacks	V112 / 3.0 MW (tip 136 m)	1 059	7 239	2 453	17.76
S3 nature	V112 / 3.0 MW (tip 136 m)	912	6 232	2 459	15.32
S4 landscape	V112 / 3.0 MW (tip 136 m)	716	4 892	2 458	12.03
S5 risk	V112 / 3.0 MW (tip 136 m)	680	4 644	2 459	11.42
S6 servitudes	V112 / 3.0 MW (tip 136 m)	463	3 164	2 467	7.80
S0 pypsa eur	NREL 2020ATB 4.0 MW (tip 185 m)	3 757	19 079	3 230	61.62
S1 zoning	NREL 2020ATB 4.0 MW (tip 185 m)	6 266	31 824	3 258	103.68
S2 setbacks	NREL 2020ATB 4.0 MW (tip 185 m)	973	4 943	3 243	16.03
S3 nature	NREL 2020ATB 4.0 MW (tip 185 m)	831	4 221	3 250	13.72
S4 landscape	NREL 2020ATB 4.0 MW (tip 185 m)	651	3 305	3 251	10.75
S5 risk	NREL 2020ATB 4.0 MW (tip 185 m)	618	3 140	3 252	10.21
S6 servitudes	NREL 2020ATB 4.0 MW (tip 185 m)	420	2 136	3 260	6.96
S0 pypsa eur	NREL 2020ATB 5.5 MW (tip 208 m)	3 757	19 275	3 302	63.65
S1 zoning	NREL 2020ATB 5.5 MW (tip 208 m)	6 266	32 150	3 332	107.12
S2 setbacks	NREL 2020ATB 5.5 MW (tip 208 m)	929	4 768	3 316	15.81
S3 nature	NREL 2020ATB 5.5 MW (tip 208 m)	790	4 051	3 323	13.46
S4 landscape	NREL 2020ATB 5.5 MW (tip 208 m)	617	3 163	3 324	10.52
S5 risk	NREL 2020ATB 5.5 MW (tip 208 m)	586	3 007	3 325	10.00
S6 servitudes	NREL 2020ATB 5.5 MW (tip 208 m)	398	2 043	3 334	6.81

Bibliographic reference for the comparison of §8.1: Clymans, W., Poelmans, L., Meyvis, L., et al. (2022), *BREGILAB WP3 RES Generation: Wind & PV deployment evolution and availability factor*, VITO/EnergyVille for the FPS Economy.

Table 14: Allocation results for every turbine class and spacing. *free* is the minimum-distance allocation alone, in row-major and pseudo-random order; *parks* adds the park rule, smaller groups included on the land a park of four cannot use; *horizon* the open-horizon rule; the *km* rows add the inter-distance between parks; one row keeps only parks of four or more.

turbine	model	spacing	d/D	order	parks	machines	P [MW]
V112 / 3.0 MW (tip 136 m)	free	observed	4.3	row major	–	4 821	14 463
	free	observed	4.3	random	–	4 156	12 468
	parks + horizon	observed	4.3		457	3 478	10 434
	free	crosswind	5.0	row major	–	4 114	12 342
	free	crosswind	5.0	random	–	3 499	10 497
	parks	crosswind	5.0		609	4 022	12 066
	parks + horizon	crosswind	5.0		470	2 934	8 802
	parks + horizon + 4km	crosswind	5.0		278	2 268	6 804
	parks + horizon + 6km	crosswind	5.0		192	1 779	5 337
	parks ≥ 4 , no exception + horizon	crosswind	5.0		197	2 534	7 602
	free	isotropic	5.9	row major	–	3 443	10 329
	free	isotropic	5.9	random	–	2 957	8 871
	parks + horizon	isotropic	5.9		475	2 451	7 353
NREL 2020ATB 4.0 MW (tip 185 m)	free	observed	4.3	row major	–	3 213	12 852
	free	observed	4.3	random	–	2 760	11 040
	parks + horizon	observed	4.3		467	2 280	9 120
	free	crosswind	5.0	row major	–	2 865	11 460
	free	crosswind	5.0	random	–	2 458	9 832
	parks	crosswind	5.0		611	2 771	11 084
	parks + horizon	crosswind	5.0		453	2 043	8 172
	parks + horizon + 4km	crosswind	5.0		284	1 579	6 316
	parks + horizon + 6km	crosswind	5.0		193	1 223	4 892
	parks ≥ 4 , no exception + horizon	crosswind	5.0		168	1 620	6 480
	free	isotropic	5.9	row major	–	2 478	9 912
	free	isotropic	5.9	random	–	2 138	8 552
	parks + horizon	isotropic	5.9		481	1 769	7 076
NREL 2020ATB 5.5 MW (tip 208 m)	free	observed	4.3	row major	–	2 737	15 054
	free	observed	4.3	random	–	2 381	13 096
	parks + horizon	observed	4.3		462	1 973	10 852
	free	crosswind	5.0	row major	–	2 373	13 052
	free	crosswind	5.0	random	–	2 083	11 456
	parks	crosswind	5.0		632	2 291	12 600
	parks + horizon	crosswind	5.0		489	1 726	9 493
	parks + horizon + 4km	crosswind	5.0		294	1 271	6 990
	parks + horizon + 6km	crosswind	5.0		201	1 013	5 572
	parks ≥ 4 , no exception + horizon	crosswind	5.0		151	1 246	6 853
	free	isotropic	5.9	row major	–	2 011	11 060
	free	isotropic	5.9	random	–	1 764	9 702
	parks + horizon	isotropic	5.9		509	1 454	7 997

Table 15: Source datasets, as retrieved.

dataset	service / layer	features	retrieved
Points d'adresse ICAR	DONNEES_BASE/ICAR_- ADR_PT/1	1 512 246	2026-09-22
Périmètres d'intérêt paysager de l'inventaire ADESA	AMENAGEMENT_TERRI- TOIRE/ADESA/1	1 142	2026-09-22
Carte d'évaluation pour les obstacles aéronautiques (DGTA)	AMENAGEMENT_TERRI- TOIRE/AOEM/0	6 435	2026-09-22
Cavités souterraines d'intérêt scientifique	FAUNE_- FLORE/CONSNAT/0	80	2026-09-22
Zones inondables — débordement, période de retour extrême	EAU/ZI/2	39 893	2026-09-22
Biens classés — ensembles architecturaux	AMENAGEMENT_TERRI- TOIRE/BC_PAT/2	53	2026-09-22
Biens classés — zones de protection	AMENAGEMENT_TERRI- TOIRE/BC_PAT/4	367	2026-09-22
Biens classés — sites classés	AMENAGEMENT_TERRI- TOIRE/BC_PAT/1	1 259	2026-09-22
Patrimoine mondial UNESCO — zones tampon	AMENAGEMENT_TER- RITOIRE/PAT_MND_UN- ESCO/1	24	2026-09-22
Zones de contraintes karstiques	AMENAGEMENT_TERRI- TOIRE/CONTR_KARST/0	449	2026-09-23
Contraintes physiques liées aux éboulements	AMENAGEMENT_TERRI- TOIRE/RISQ_EBOULT/0	997	2026-09-22
Réseau Natura 2000 — périmètres des sites en vigueur	FAUNE_- FLORE/NATURA2000/9	240	2026-09-22
Plan de secteur — lignes électriques haute tension	AMENAGEMENT_TERRI- TOIRE/PDS/12	1 016	2026-09-22
Plan de secteur — périmètres d'intérêt paysager	AMENAGEMENT_TERRI- TOIRE/PDS/17	1 456	2026-09-22
Plan de secteur — réseau ferroviaire	AMENAGEMENT_TERRI- TOIRE/PDS/11	911	2026-09-22
Plan de secteur — réseau routier	AMENAGEMENT_TERRI- TOIRE/PDS/9	5 420	2026-09-22
Plan de secteur — voies navigables	AMENAGEMENT_TERRI- TOIRE/PDS/13	1	2026-09-22
Plan de secteur — zones d'affectation	AMENAGEMENT_TERRI- TOIRE/PDS/22	43 797	2026-09-22
Réserves naturelles agréées	FAUNE_- FLORE/CONSNAT/3	264	2026-09-22
Réserves naturelles domaniales	FAUNE_- FLORE/CONSNAT/2	306	2026-09-22
Réserves forestières	FAUNE_- FLORE/CONSNAT/1	20	2026-09-22
Versants supérieurs à 30 degrés	AMENAGEMENT_TERRI- TOIRE/RISQ_EBOULT/1	2 834	2026-09-22
Zones de prévention rapprochée (IIa) des captages	EAU/PROTECT_CAPT/2	736	2026-09-22
Zones humides d'intérêt biologique	FAUNE_- FLORE/CONSNAT/4	73	2026-09-22